AMITY SCHOOL OF ENGINEERING & TECHNOLOGY

PROJECT PROGRESS REPORT

B.Tech (Computer Science and Engineering - AIML) | 4th Year / 7th Semester
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Group No.: 50

Project Title: GuardShield AI: Real-Time Sidecar Proxy for LLM Hallucination Detection & Prompt Injection Defense

Project Area: Generative AI & AI Safety (Trustworthy AI / MLOps)

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Project Team Details:

Programme: B.TECH CSE (AIML)		**Semester: 7th Semester (Odd 2026-2027)**	
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1. Executive Summary & Problem Formulation

Modern Large Language Models (LLMs) deployed across high-stakes domains (healthcare, legal counsel, autonomous agent workflows, and financial services) suffer from two compounding, catastrophic failure modes: (1) Adversarial Ingress Attacks, where malicious users craft direct prompt injections, jailbreaks, and obfuscated payload wrappers that bypass system instructions; and (2) Hallucinatory Egress Generation, where autoregressive models emit plausible-sounding but factually ungrounded assertions with high predictive confidence.

Existing commercial safety guardrails exhibit severe operational trade-offs. Heavy LLM-based classifiers such as Meta Llama-Guard-3 (8B parameters) impose between 1,500 ms and 3,500 ms of latency per query, rendering them unfeasible for sub-second production microservices. Conversely, post-generation factual consistency detectors (e.g., SelfCheckGPT) rely on stochastic multi-sample generation, compounding inference costs and introducing multi-second offline bottlenecks. GuardShield AI introduces a high-throughput, low-latency dual-stage sidecar proxy architecture combining: (a) a fine-tuned DeBERTa-v3-small Pre-Scan Filter (44.3M backbone, 142.6M total params) intercepting prompt injections in ~13.09 ms (p50) with sliding-window chunking; (b) a Cross-Encoder Natural Language Inference (NLI) Verifier performing premise-

hypothesis fact grounding in ~13.89 ms (p50); and (c) a dynamic streaming Shannon entropy engine halting generation mid-sentence when sustained uncertainty spikes occur.

2. Comprehensive Literature Review & Research Gaps

A systematic survey of 25 peer-reviewed publications across premier AI conferences (NeurIPS, EMNLP, ACL, NAACL, and ICLR) establishes the theoretical foundation and empirical gap for GuardShield AI:

• Adversarial Jailbreaking & Prompt Injection: Chao et al. [1] (JailbreakBench, NeurIPS 2024) and Schulhoff et al. [2] (HackAPrompt, EMNLP 2023) demonstrated that token-level adversarial perturbations and semantic overrides consistently breach commercial LLM alignment. Ji et al. [3] (BeaverTails, NeurIPS 2023) and Zhang et al. [4] (CAPTURE, ACL 2025) highlighted the necessity of semantic intent classification rather than superficial regex heuristics. Inan et al. [5] (Llama-Guard, Meta AI 2024) proposed heavy guardrail models that cannot satisfy sub-25ms microservice latency budgets.

• Hallucination Detection & Fact Grounding: Manakul et al. [6] (SelfCheckGPT, EMNLP 2023) formulated sampling-based hallucination detection but incurred heavy computational overhead. Li et al. [7] (HaluEval, EMNLP 2023) and Thorne et al. [8] (FEVER, NAACL 2018) established rigorous benchmarks for factual verification. Lin et al. [9] (TruthfulQA, ACL 2022) revealed that LLMs frequently imitate human misconceptions unless anchored to explicit premise context.

• Compact Representation Learning & NLI: He et al. [10] (DeBERTa-v3, ICLR 2023) demonstrated that disentangled attention and ELECTRA-style pre-training deliver state-of-the-art NLU performance while remaining exceptionally compact. Reimers & Gurevych [11] (Cross-Encoder, EMNLP 2019) proved that joint premise-hypothesis cross-attention dramatically outperforms bi-encoder cosine similarity for entailment.

• Dynamic Token Logit Entropy: Shannon [12] (1948) established discrete entropy formulation. Azaria & Mitchell [13] (EMNLP 2023) and Kuhn et al. [14] (Semantic Entropy, ICLR 2023) demonstrated that internal logit distributions and token uncertainty directly correlate with hallucination drift.

Identified Research Gap: Existing solutions force an unviable choice between high-latency multi-second LLM evaluators and fragile keyword filters. GuardShield AI fills this critical void by engineering an ultra-compact (~142M param) dual-stage sidecar pipeline delivering 98.42% security accuracy and sub-70ms end-to-end sequential latency.

3. GuardShield AI Architecture & Technical Implementation

GuardShield AI operates as an asynchronous sidecar microservice positioned between end-users and target generative LLMs (e.g., Qwen2.5-3B, TinyLlama-1.1B, Llama-3-8B):

1. Stage 1 (Pre-Scan Security Filter): Implemented via fine-tuned microsoft/deberta-v3-small. Ingress prompts are processed using sliding-window chunking (max 126 tokens, stride 64 tokens) to prevent adversarial length-padding bypasses. Prompts yielding injection probability P >= 0.50 are terminated immediately at the gateway with an HTTP 403 Forbidden status, avoiding 100% of downstream LLM generation compute.

2. Dynamic Token Logit Entropy Engine: For approved prompts, as the generative LLM emits tokens, the engine computes normalized conditional Shannon entropy over the top-k distribution: H(x) = -sum(p * log2(p)) in [0, log2(k)]. A calibrated dual-condition threshold (H >= 1.35 bits and Top-1 probability < 0.50) isolates genuine hallucination drift from confident tokens (e.g., 'Paris' with p=0.62-0.99 yields H<0.5 bits, status OK; while speculative hallucinations drop to p=0.35 with H=1.58 bits, triggering early termination after 2 consecutive spikes).

3. Stage 2 (Cross-Encoder NLI Fact Grounding): Candidate output assertions are evaluated against authoritative retrieval context using cross-encoder/nli-deberta-v3-small. To eliminate length and syntactic confounding where short entity

answers artificially skew predictions, QA pairs are reformulated into complete declarative statements ('The answer to the question {Q} is {A}.'), ensuring robust semantic entailment verification.

4. Empirical Evaluation & Multi-Benchmark Research Results (5,948 Samples)

GuardShield AI was empirically benchmarked across 7 peer-reviewed academic datasets totaling 5,948 curated samples. In accordance with strict scientific standards, all evaluation scripts operate on raw model logits with zero keyword heuristics. 95% Wilson score binomial confidence intervals and length-matched scientific controls were established to guarantee publication validity.

Evaluation Benchmark	Dataset Source(s)	Sample Count	Accuracy (%)	Recall (%)	Precision (%)	FPR (%)	Median Latency
Stage 1 In-Distribution Test	Stanford SPML + Jayavibhav + Deepset	698 held-out	98.42% ± 0.96%	97.75% (347/355)	99.14%	0.87% (3/343)	32.95 ms (p50)
Stage 1 Zero-Shot OOD Generalization	ChatGPT JailbreakBench + Held-out Benign	500 unseen	98.20% ± 1.22%	97.60% (244/250)	98.79%	1.20% (3/250)	18.48 ms (p50)
Scientific Length-Matched Control	Length-Controlled Balanced Split	198 control	98.48% (Length: 60.61%)	96.97%	100.00%	0.00%	24.12 ms (p50)
Long Safe Prompt False-Positive Audit	Academic Technical Queries (>= 200 chars)	66 long prompts	95.45% Pass Rate	--	--	4.55% (3/66)	36.65 ms (p50)
Stage 2 Authentic Fact Grounding	FEVER (NAACL) + HaluEval (EMNLP)	800 unique pairs	73.25% ± 3.06%	66.50% (FEVER: 88.5%)	76.88%	20.00%	18.91 ms (p50)
Sequential End-to-End Pipeline (CUDA)	Combined Stage 1 + Stage 2 Sequential	100 real queries	--	--	--	--	65.86 ms (p50) Mean: 68.47 ms

5. Cross-Model Defense Efficiency & Compute Savings Projection

By intercepting malicious prompts in ~13.09 ms (p50) at Stage 1, GuardShield AI eliminates the need to execute full multi-token autoregressive decoding on downstream generative LLMs. For a standard 50-token response cycle across commodity foundation models:

• GPT-2 (124M): Unprotected latency = 175.4 ms vs GuardShield = 12.87 ms (92.66% latency reduction; 48 tokens saved).
• TinyLlama-1.1B: Unprotected latency = 740.2 ms vs GuardShield = 12.87 ms (98.26% latency reduction; 65 tokens saved).
• Qwen2.5-3B-Instruct: Unprotected latency = 1,850.6 ms vs GuardShield = 12.87 ms (99.30% latency reduction; 92 tokens saved).
• Qwen2.5-7B-Instruct: Unprotected latency = 3,420.0 ms vs GuardShield = 12.87 ms (99.62% latency reduction; 120 tokens saved).

In total, GuardShield avoids 100% of the GPU memory allocation, KV-cache management, and FLOPs for intercepted attacks, shielding enterprise infrastructure from compute exhaustion and denial-of-wallet (DoW) exploits.

6. PERT Schedule & Milestone Progress (Weeks 1 to 12)

Milestone / Task Description	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
1. Literature Review & Research Gap Formulation	✓	✓	✓									
2. Architecture & Environment Setup (CUDA, PyTorch 2.6)	✓	✓										
3. Synthetic Ingestion & Preliminary SLM Fine-Tuning		✓	✓	✓								
4. Streaming Token Logit Shannon Entropy Engine				✓	✓	✓						
5. Cross-Encoder NLI Fact Grounding Pipeline					✓	✓	✓					
6. FastAPI Async Proxy & Concurrency Isolation						✓	✓	✓				
7. Mega Benchmark Ingestion (5,948 samples across 7 datasets)							✓	✓	✓			
8. In-Distribution & Zero-Shot OOD Generalization Testing								✓	✓			
9. Authentic FEVER/HaluEval Grounding & Wilson CIs									✓	✓		
10. Empirical E2E Latency Profiling & Live Colab Suite										✓		
11. IEEE/ACM Research Paper Manuscript										✓	✓	

Drafting												
12. Capstone Final Viva, Presentation Slides & Archive											✓	✓

7. Summary of Phase Achievements (Week 10 Review)

1. Full Provenance & Authentic Data: Replaced preliminary synthetic pairs with 800 unique, published benchmark pairs from FEVER (NAACL 2018) and HaluEval (EMNLP 2023). Declarative hypothesis reformulation eliminated length confounders, achieving 86.50% accuracy on FEVER and 60.00% on HaluEval (73.25% overall, 76.88% precision).

2. Statistical Rigor & Controls: Integrated 95% Wilson score confidence intervals across all splits (In-Dist: 98.42% ± 0.96%, OOD: 98.20% ± 1.22%). Executed length-matched control experiment (N=198), proving GuardShield maintains 98.48% accuracy while length heuristic collapses to 60.61%.

3. Real Hardware Sequential Latency: Established true empirical sequential E2E defense overhead of 65.86 ms (p50) / 68.47 ms (mean) across 100 real CUDA queries, logged per-sample in raw_eval_pipeline_e2e.json.

4. Mathematical Streaming Entropy Calibration: Calibrated normalized top-k Shannon entropy with a dual-condition threshold (H >= 1.35 bits and Top-1 < 0.50), preventing false early terminations on confident tokens while halting genuine hallucination drift mid-sentence.

5. Interactive Multi-Model Demonstration Suite: Enhanced Google Colab notebook with interactive custom prompt fields, natural conversational presets, chat template formatting, and native support for modern instruction LLMs (Qwen2.5-3B, TinyLlama-1.1B, Qwen2.5-1.5B).

8. Peer-Reviewed Academic References

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